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TowMagKit: A hybrid Python-Fortran toolkit for marine towed total-field magnetometer processing

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Summary

TowMagKit (Marine towed total-field magnetometry toolkit) is a Python-centric package that consolidates every major stage of marine towed total-field magnetometer (TFM) processing into a single, scriptable workflow. The front-end driver 'scripts/run-raw2corrected.py' converts raw sensor logs into intermediate ASCII files, applies moving-average smoothing, performs cable-layback correction to reconcile the spatial offset between the magnetometer and the vessel's GPS antenna, executes International Geomagnetic Reference Field (IGRF) (International Association of Geomagnetism and Aeronomy, 2024) subtraction via IAGA-VMOD/ppigrf (Laundal et al., 2024), and prepares observatory records for diurnal-variation correction before applying the correction. After diurnal effects have been removed, the data are partitioned into straight-line survey legs (Douglas & Peucker, 1973; Ramer, 1972) and turning-motion intervals to facilitate line-based anomaly analysis. Crossover levelling is then performed by 'scripts/run-crossover.py', which wraps legacy Fortran routines in thin Python bindings to implement the Ishihara algorithm (Ishihara, 2015). The resulting outputs remain fully compatible with GMT's 'x2sys' (Wessel, 2010; Wessel et al., 2019); definition files and a Bash pipeline for batch processing reside in the 'gmt_scripts/' directory, alongside example scripts for gridding the final anomalies.

Statement of Need

Traditional TFM processing has relied on a scattered mix of Bash/Fortran/C/MATLAB scripts and executables, making the execution order opaque, reproducibility was fragile, and modification cumbersome. TowMagKit replaces that mix with a single, readable codebase, where top-level 'run-*.py' drivers lay out the entire sequence of steps, so users can follow the flow line-by-line and preserve intermediate outputs to guarantee reproducibility. At the same time, the numerical routines reside in a modular API under 'src/', allowing algorithms to be swapped or extended without interacting the rest of the pipeline. This arrangement lowers the entry barrier for newcomers and keeps the workflow transparent and maintainable.

Documentation and Examples

A key feature of TowMagKit is that it represents the first open-source packaging of the Ishihara crossover-adjustment algorithm (Ishihara, 2015) for the Python ecosystem. Instead of re-implemis wrapped enting the routine from scratch, the original Fortran code in thin Python bindings, preserving its bit-level behavior while making it directly callable from modern workflows. The method minimizes line-to-line misfits not only at conventional crossover points

but also in surveys where crossovers are sparse or unreliable. It explicitly models the combined effects of external-field disturbances, hull magnetization, and navigation errors, iteratively solving for per-track level shifts that yield a self-consistent data set. This makes the algorithm especially valuable when observatory-based diurnal correction is impractical—e.g. in remote oceans far from fixed stations—or when CM4-based estimates of diurnal and secular variation carry large uncertainties (Oda, 2022). Its proven robustness led to adoption for integrating marine data into the second World Digital Magnetic Anomaly Map (WDMAM v2) (Lesur et al., 2016).

'./examples/GS24/'

This project demonstrates an end-to-end workflow for the GS24 cruise that used a Cesium magnetometer (G-880 Geometrics Inc.). It includes raw '.txt' logs, all intermediates produced by 'run-raw2corrected.py' and 'run-crossover.py', and shell scripts for optional GMT gridding. Because licensing prevents redistribution of observatory data, users must download the one-minute Kakioka files (Kakioka Magnetic Observatory, 2013) themselves and place the '.min' files in 'examples/GS24/dv/' before running the pipeline.

'./examples/Hakuho/'

This example provides a minimal sample of raw data recorded by a surface proton precession magnetometer (PR745, Kawasaki Geological Engineering Co., Ltd.) during the KH-22-10 cruise (Koge, 2022), for use in quick file-conversion tests.

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